CSCI 3141: Foundations of Data Science in R *(and Python)*

Course Syllabus

Section A. Course Information

A.1 Administrative information

Table 1. Course and instructor information

Field	Course information
Delivery	In person, with practical work during confirmed class meetings
Location	Carleton Tupper Building, #3H01 (Tupper 3H01)
Course website	MAPS-Lab/CSCI-3141 (public course materials)
Meetings	Tue and Thu, 16:05 to 17:25
Instructor	Prof. Gabriel Spadon, PhD
Email	spadon@dal.ca
Office hours	By appointment through Microsoft Teams

Email private questions; use Teams for class-wide questions. Replies in three business days.

A.2 Important dates

Table 2. Term dates and their effect on the course

Date	Event	Course effect
Sep 8	Fall classes begin	First CSCI 3141 meeting
Sep 22	Last day to add or change classes	Student advising reminder
Sep 30	National Day for Truth and Reconciliation	University closed; no Tue or Thu meeting lost
Oct 7	Last day to drop without W	Student advising reminder
Oct 12	Thanksgiving Day	University closed; no Tue or Thu meeting lost
Oct 15	Midterm revision meeting	Office hours in class, no new content
Oct 20	Midterm examination	In class, 16:05 to 17:25
Nov 5	Last day to drop with W	Student advising reminder
Nov 9 to 13	Fall Break	No meetings or new required activity; Assignment 2 and Lab 09 fall due after it
Dec 3	Final regular CSCI 3141 meeting	Cumulative review and workflow synthesis
Dec 8	Monday schedule held	No CSCI 3141 meeting; Lab 12 due at 16:05
Dec 9	Wednesday schedule; classes end	Assignment 3 due at 23:59; no class
Dec 11 to 20	Formal examination period	Final details set by the Registrar

> Wait for the Registrar's final examination schedule before making travel plans.

A.3 Course description and prerequisites

This course moves from data import and preparation through visualization, modelling, and reporting. R is primary, with full Python parity in materials, demonstrations, practical work, and assessments. No formal prerequisite course is listed; expected preparation appears below.

A.4 Course rationale and scope

The course follows a complete data science workflow, with emphasis on reproducibility, interpretation, uncertainty, and evidence-based modelling choices.

Who this course is for

Students from any major are welcome. Expected preparation includes calculus, statistics, basic computing, and programming experience. Guided examples introduce the course tools before students use them in assessed work.

A.5 Learning outcomes

By the end of the course, you should be able to meet each outcome below.

- Describe the main steps and decision points in a data science project.
- Import, clean, manipulate, transform, and document data in preparation for analysis.
- Use descriptive statistics and appropriate visualizations in R or Python.
- Explain and apply the predictive modelling methods presented in the course.
- Explain and apply clustering, similarity, distance, and outlier-detection methods.
- Evaluate, compare, and select models for a dataset.
- Prepare reproducible analyses and reports in R Markdown or Python notebooks.

A.6 Delivery, communication, resources, and computing

We meet in person on Tuesday and Thursday from 16:05 to 17:25 in Carleton Tupper Building 3H01. Practical work takes place during confirmed Thursday meetings, and in Week 12 during the Tuesday meeting.

Bring a laptop to practical meetings. If computer access is a barrier, contact me early so we can arrange an equivalent way to participate.

The public course repository holds announcements, materials, assessment instructions, and submission requirements. Reading course materials does not require a GitHub account. You need an account for your assessment portfolio. Check the repository and your Dalhousie email regularly. Grades and feedback remain private.

Texts, software, and data There is no required paid textbook. Readings and software guidance are provided through the course repository. Use a local R or Python environment, or a free cloud notebook environment such as one of these three.

- Posit Cloud, with RStudio or Jupyter projects for R or Python;
- Kaggle Notebooks, with R or Python notebooks; or
- Google Colab, with Python notebooks.

Choose the option that best suits your language and workflow, or another free cloud notebook service. Keep your notebooks and supporting files in your GitHub portfolio. No paid external service or public generative AI service is required.

Course datasets will be supplied or linked with access instructions.

When using Faculty of Computer Science computing resources, follow the Dalhousie Acceptable Use Policy and the Faculty's Responsible Computing Policy.

- **Acceptable Use Policy** (link)
- **Responsible Computing Policy** (link)

A.6.1 Building and preserving your portfolio

Labs and assignments build your public data-science portfolio for class discussion after release. Keep assessed work private before its deadline, including code, reports, outputs, cloud notebooks, and answer-revealing links. A branch in a public repository is public too.

1. Work in a private GitHub repository with instructor access, under `labs/lab-XX/` or `assignments/assignment-XX/`. Commit and push regularly as you work.
2. Commit the completed work, then record its full SHA and immutable commit URL in that directory's `README.md` in a second commit. Push both by the deadline. The marker assesses the recorded snapshot; branch links and movable tags are insufficient. Keep the work commit reachable. The pushed record is deadline evidence; Git author dates are not.
3. After the deadline and instructor release, check all files and reachable history. If another assessment is unreleased, keep the repository private until instructor authorization; never rewrite history to remove it. Otherwise publish the same submitted commit and history in your public GitHub portfolio, preserving its SHA when merging. Present that version at the next class meeting, if one follows; label later improvements separately.

Preserve history permanently. Never amend, rebase, squash, delete, replace, or force-push recorded assessment commits, move submitted tags, or fabricate or backdate progress. Correct through new commits; contact the instructor for an approved correction of accidentally committed private information. Git records contents, not proof of authorship or protection against rewriting. The recorded submission SHA and instructor-retained snapshot are the independent assessment reference.

Grades and feedback remain private. Do not publish personal or instructor-marked private material. When accommodation, safety, privacy, copyright, or licensing requires it, use an equivalent private GitHub repository with the same evidence and rubric instead of publishing a public portfolio.

A.7 Assessment and grading

Your numerical grade combines weekly practice, three assignments, and two examinations. Assessment completion requirements also apply as described below.

Table 3. Assessment components and their weights

Component	Weight	Core evidence
Integrated labs	15%	Best 9 of 10 individual lab results
Assignments	45%	Three individual data science analyses
Midterm examination	15%	In-class independent examination
Final examination	25%	Registrar-scheduled cumulative examination
In-class quizzes	Ungraded	Completion requirement in A.7.4

A.7.1 Integrated labs

Each week's second class begins an integrated lab, except in Weeks 06 and 07, which carry no lab. Lab 12 begins with Week 12's first class, so that it falls due before classes end. Choose an approved dataset and work in R or Python with equal requirements, feedback, and grading.

Prepare each lab in `labs/lab-XX/`; submit its fixed commit under A.6.1 seven calendar days after release, before 16:05 Halifax time (the usual Class B start), even in weeks without class. Lab 09 is the one exception and falls due at the first Class B after Fall Break, November 19. Best 9 of the 10 labs count toward the 15% lab component.

Week 1 establishes observational units, variable types, descriptive summaries, plots, and the private submission workflow. Model examples in the orientation are previews; Lab 01 does not require model fitting or significance tests.

A.7.2 Assignments

Table 4. Assignment release dates, due dates, weights, and focus

Item	Release	Due at 23:59	Weight	Primary focus
Assignment 1	Sep 29	Oct 20	7.5%	Import and basic exploration
Assignment 2	Oct 27	Nov 17	7.5%	Predict fatal aviation accidents from the Assignment 1 table
Assignment 3	Nov 19	Dec 9	30%	Descriptive modelling solution

Assignments are individual work. Prepare each privately under `assignments/assignment-XX/`. Submit the fixed commit privately and publish it only after release, following A.6.1. The required files must open and reproduce the analysis. Assignment instructions provide the rubric, dataset, format, and authorized tools but do not change the published weights without an approved course notice.

A.7.3 Late work and absence

Assignments 1 and 2 lose 5 percentage points per late calendar day for up to seven days. Work after that window is accepted only through an approved accommodation or absence process. Assignment 3 has no routine late window.

For an eligible short-term absence of up to three consecutive days, contact me before the missed requirement and submit a Student Declaration of Absence within three calendar days after the last day of the absence. You may use the declaration at most twice per course. Follow the Student Absence Regulation for longer absences. Contact me about a missed midterm; final examinations follow the Registrar's separate process.

The lab drop can absorb a missed item. When it does not resolve an eligible absence, an equivalent alternative or denominator adjustment applies.

A.7.4 In-class assessments and examinations

Four individual in-class quizzes take place during the term, two before the midterm and two after it. Their dates are selected in advance by the instructor and are not announced. Each quiz is recorded as completed or not completed; quiz marks do not increase or decrease the weighted course percentage. Students must complete at least two of the four quizzes (50%) to pass the course, subject to the absence provisions below.

A missed quiz is recorded as not completed unless the absence is covered by a valid Student Declaration of Absence, accepted medical documentation, or an approved accommodation under A.7.3.

Approved absences carry no penalty; an equivalent completion opportunity or an adjusted completion requirement will be provided. Students who do not meet the completion requirement after these provisions have been applied cannot pass the course, regardless of their weighted course percentage.

The midterm is in class on October 20 from 16:05 to 17:25. The Registrar schedules the cumulative final examination. Bring valid Dalhousie photo ID.

Course and personal notes are allowed unless an examination lists narrower approved aids. Internet access, communication, and generative AI are prohibited.

A.7.5 Passing requirements and grade scale

Passing requirements

- The weighted course result must be at least 50 percent.
- The combined assignment component must be at least 50 percent.
- The final examination must be at least 50 percent.
- The assessment completion requirement in A.7.4 must be met, with approved absence and accommodation provisions applied.
- A minimum grade of C is required when this course is core to an FCS degree or is used as a prerequisite for a subsequent CSCI course.

Table 5. Letter grade scale

Letter	Range	Letter	Range
A+	90 to 100	B+	77 to 79
A	85 to 89	B	73 to 76
A-	80 to 84	B-	70 to 72
C+	65 to 69	D	50 to 54
C	60 to 64	F	0 to 49
C-	55 to 59		

A.8 Tentative topic and assessment schedule

The weekly plan may change in pace or examples. Any update will be posted in the course repository and remain consistent with the approved course and assessment rules.

Table 6. Weekly topics and assessment milestones

Week	Dates	Topics	Milestones
01	Sep 8 Sep 10	LEC-01 Course orientation and the data science process LEC-02 Summarizing data	Lab 01
02	Sep 15 Sep 17	LEC-03 Data preprocessing foundations LEC-04 Cleaning, transformation, and feature engineering	Lab 02
03	Sep 22 Sep 24	LEC-05 Preprocessing and visualizing data LEC-06 Visualizing data	Lab 03

Continued on next page

Week	Dates	Topics	Milestones
04	Sep 29 Oct 1	LEC-07 Manipulation and visualization in R and Python LEC-08 Reporting in R Markdown and Python notebooks	Lab 04 A1 release
05	Oct 6 Oct 8	LEC-09 Predictive analytics and classification metrics LEC-10 Predictive analytics and regression metrics	Lab 05
06	Oct 13 Oct 15	LEC-11 KNN, linear regression, and LDA Midterm revision in office hours	
07	Oct 20 Oct 22	Midterm examination LEC-12 Performance estimation methods	A1 due
08	Oct 27 Oct 29	LEC-13 Performance estimation continued LEC-14 Support vector machines	Lab 08 A2 release
09	Nov 3 Nov 5	LEC-15 Decision trees introduction LEC-16 Tree-based models and pruning	Lab 09
10	Nov 17 Nov 19	LEC-17 Bias, variance, overfitting, and regularization LEC-18 Descriptive analytics, similarity, and distance	Lab 10, A2 due A3 release
11	Nov 24 Nov 26	LEC-19 Distance and clustering preparation LEC-20 Clustering algorithms and evaluation	Lab 11
12	Dec 1 Dec 3	LEC-21 Outlier detection LEC-22 Cumulative review and workflow synthesis	Lab 12 A3 due Dec 9

A.9 Collaboration, generative AI, and academic integrity

Discussing course ideas is encouraged, but submitted work must show your own understanding. Do not share answers, code, analysis files, answer-producing prompts, or solution steps unless written instructions authorize collaboration.

Table 7. Permitted and prohibited generative AI use by context

Context	Status	Requirement
Private study planning and practice	Permitted	Check outputs against course sources and protect private or copyrighted information
Editing a student-written ungraded note	Permitted	The intellectual work must already be the student's
A named use authorized in assignment instructions	Permitted only as stated	Disclose the tool, purpose, interaction summary, and changes made
Integrated lab submission	Prohibited	Generated code, prose, analysis, or figures cannot be submitted for lab credit
Assignments	Prohibited unless expressly authorized	Silence is not permission
In-class quizzes, midterm, and final examinations	Prohibited	Complete the work independently
Uploading non-public course or student material	Prohibited	Protect privacy, copyright, intellectual property, and assessment security

No paid or public generative AI service is required. Unauthorized use on assessed work enters the academic integrity process. Authorized uses and disclosure requirements appear in the assessment instructions.

A.9.1 Academic integrity and originality checking

Submit work that can honestly be presented as your own. Plagiarism, unauthorized collaboration or AI use, fabrication, reused credit, and helping another person commit an offence are prohibited.

I may review code and writing with Moss, another approved service, or direct source comparison. A match prompts review and does not determine a grade.

- **Academic integrity** (link)
- **Originality checking policy** (link)

A.10 Course-specific policies

A.10.1 Attendance, participation, and access

Attendance is not graded, but you remain responsible for missed material and announcements. Use the applicable absence or accommodation process when needed.

A.10.2 Recording

Some classes may be recorded and shared privately, but recordings are not guaranteed. Do not record, photograph, stream, or redistribute a class without written permission or an accommodation. Contact me for a privacy-safe way to participate.

A.10.3 Course materials and copyright

Unless the instructor explicitly marks material private or restricted, course materials released by the instructor may be shared with credit to the author and CSCI 3141. This includes the publicly released syllabus, slides, notes, and assessment handouts. Keep source links, third-party attributions, and applicable licence terms. Copyright remains with the respective owners.

This permission does not release student work before its submission deadline or authorize sharing another student's work. Follow A.6.1 for the assessed portfolio. Keep instructor-marked private material, grades, feedback, and unreleased recordings private.

Section B. University Statements

B.1 Territorial acknowledgement

Dalhousie University operates in the unceded territories of the Mi'kmaw, Wolastoqey, and Peskotomuhkati Peoples. These sovereign nations hold inherent rights as the original peoples of these lands, and we each carry collective obligations under the Peace and Friendship Treaties. Section 35 of the Constitution Act, 1982 recognizes and affirms Aboriginal and Treaty rights in Canada.

We recognize that African Nova Scotians are a distinct people whose histories, legacies and contributions have enriched that part of Mi'kma'ki known as Nova Scotia for over 400 years.

B.2 Internationalization

Data science decisions affect communities beyond where a model is built. Our work will respect cross-cultural knowledge and consider global consequences.

B.3 Academic integrity

Dalhousie's values of honesty, trust, fairness, responsibility, and respect apply throughout the term. Section A.9 explains the course requirements.

B.4 Accessibility and accommodation

For barriers in course design, instruction, technology, or assessment, contact the Student Accessibility Centre or speak with me. Accommodation information is private and follows the approved University process.

B.5 Culture of respect, diversity, and inclusion

Everyone shares responsibility for a respectful, inclusive, and scent-aware learning environment. Academic disagreement must remain civil and free from harassment or discrimination.

The Faculty of Computer Science Culture of Respect initiative, CoReCS, offers resources for building a welcoming, safe, and respectful community.

- **CoReCS** (link)

B.6 Student conduct

Everyone has the right to be respected and safe. The Code of Student Conduct and other University processes apply when those expectations are not met.

B.7 Fair dealing and copyright

The Fair Dealing Policy governs limited use of copyrighted material. Follow the applicable licence terms for third-party content. Section A.10.3 authorizes sharing instructor-released course materials with credit and explains the privacy restrictions.

B.8 Originality checking

Section A.9.1 explains approved originality checks. A match prompts review and never sets a grade.

Section C. University Policies, Guidelines, and Resources

C.1 Governing academic sources

The University Calendar and Senate rules govern this course. Use these official sources for current policies and schedules.

- **University Calendar** (link)
- **Important dates** (link)
- **Academic integrity** (link)
- **Originality checking policy** (link)
- **Student Code of Conduct** (link)
- **Fair Dealing Policy** (link)
- **Student Declaration of Absence** (link)

C.2 Student support and Faculty resources

Support is available through academic advising, Student Health and Wellness, student centres, Dalhousie Libraries, the Writing Centre, tutoring, Student Advocacy, the Ombudsperson, and Human Rights and Equity Services.

For Faculty-specific help, visit the Student Learning Centre on the second floor of the Goldberg Computer Science Building.

- **Accessibility** (link)
- **Student support** (link)
- **Student Health and Wellness** (link)